

Javascript

DOM, es6+

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>DOM</title>
</head>
<body>
  <h2>DOM </h2>

  <h3 id="element">GET ELEMENT BY ID</h3>

  <p class="class1"></p>

  <span class="class1"></span>
  <div class="class1"></div>

  <p></p>
  <p></p>
  <p></p>

  <script>
    const tagName = document.getElementsByTagName('p');

    console.log(tagName);

    console.log(tagName.length);

    for(let i=0; i<tagName.length; i++){
      tagName[i].textContent = `element ${i}`;
    }

    const divElement = document.createElement("div");
    console.log(divElement);

    divElement.textContent = "Dynamic div element";

    document.body.appendChild(divElement);
```

```
const classElement = document.getElementsByClassName("class1");
console.log(classElement[0]);
console.log(classElement[1]);
console.log(classElement[2]);

classElement[0].innerText = "hello";
classElement[1].innerText = "span";
classElement[2].innerText = "div";

classElement[0].style.backgroundColor = "blue";

const h3Element = document.getElementById("element");
console.log(h3Element);
console.log(h3Element.innerText);

h3Element.innerText = "ID";

document.write("<h1>FIIT</h1>");

// console.log("Hello there");

// {
//     let data = 10;
// }

// // console.log(data);

// const name = "FIIT";
// const age = 25;

// console.log(`name is ${name}`);
// and age is ${age}`);

// function greet({namee, agee}){
//     // console.log(obj);

//     console.log(name, age);
// }

// let object = {
//     namee: "Javascript",
//     agee : 55
```

```
// }

// greet(object);

// const {nameee } = object;

// console.log(nameee);
// // console.log(agee);


// console.log(object.name);
// console.log(object.age);


// const arr =[1,2,3,5];

// console.log(arr[0]);

// const [first , second ] = arr;

// console.log(first);
// console.log(second);


// const arr1 =[1,2,3];

// const arr2 = [4,5,6];

// const arr3 = [...arr1, ...arr2];

// console.log(arr3);


// let obj1 = {
//     a:1,
//     b:2
// }
// let obj2= {
//     c:3,
//     d:4
// }

// let obj3 = {...obj1, ...obj2};

// console.log(obj3);
```

```

    // function display(...args){
    //     console.log(...args);
    //     // console.log(a, b);
    // }

    // display(1,2 ,3, 4, 5,6 ,7,8);

    // DOM => document object model

</script>
<!-- <script src="./App.js" ></script> -->

</body>
</html>

```

Theory

1. DOM Basics

- **DOM (Document Object Model)** → a tree-like structure representing HTML elements.
- JavaScript can **select, modify, add, or remove** elements dynamically.

2. Element Selection

- **By ID** → `document.getElementById("id")`.
- **By Class** → `document.getElementsByClassName("class")`.
- **By Tag** → `document.getElementsByTagName("p")`.
- **Query Selector** → `document.querySelector(".class")` or `document.querySelectorAll("p")`.

3. Element Manipulation

- **Text content** → `element.innerText` / `element.textContent`.
- **Style changes** → `element.style.property = "value"`.
- **Attributes** → `element.setAttribute("attr", "value")`.

4. Dynamic Elements

- **Create element** → `document.createElement("div")`.
- **Append element** → `parent.appendChild(newElement)`.

5. Modern JS Features (your commented code)

- **Template literals** → ``Hello ${name}``.
- **Object destructuring** → `const {nameee} = object;`.
- **Array destructuring** → `const [first, second] = arr;`.
- **Spread operator** → `const arr3 = [...arr1, ...arr2];`.
- **Rest parameters** → `function display(...args){}`.

Coding Round Questions

- DOM basics → What is DOM and why is it important?
- Element selection → Show how to select elements by ID, class, and tag.
- Manipulation → Change text and style of a selected element.
- Dynamic elements → Create and append a new element dynamically.
- Destructuring → Explain object and array destructuring with examples.
- Spread operator → Show how to merge arrays and objects using spread.

Machine Coding Round Challenge

Problem: Dynamic Todo List

Requirements:

1. Create an input box and button.
2. On button click, add a new `<li>` to a `<ul>` list.
3. Each `<li>` should have a delete button.
4. Use DOM methods (`createElement`, `appendChild`, `removeChild`).
5. Style the list items dynamically.